

Parvez Akhtar

+91 888-232-7732 | parvezakhtar218@gmail.com | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

AI Engineer with 5+ years building and deploying production-ready ML systems and, more recently, multi-agent Agentic AI workflows that reason, plan, use tools, and retain memory. Hands-on with LLM fine-tuning (LoRA/QLoRA, PEFT), Retrieval-Augmented Generation (RAG), vector search, and rigorous LLM evaluation. Built multimodal document-intelligence pipelines that extract clean structured data at scale, and cut inference costs ~90% by taking experimental AI to reliable, cost-efficient production.

TECHNICAL SKILLS

Agentic AI: Multi-Agent Workflows & Orchestration (LangGraph, OpenAI Agents SDK, Anthropic SDK, LangChain), Reasoning / Planning / Reflection, Tool & Function Calling, Agent Memory & Human-in-the-Loop, MCP (Model Context Protocol), Structured Outputs, Guardrails & Prompt-Injection Defense, Swarm-based Orchestration.

ML & LLMs: Fine-tuning (LoRA/QLoRA, PEFT, Hugging Face), Transformer Architectures, Attention, Tokenization, Embeddings, Context-Window Optimization, Multimodal Models, Prompt Engineering, Classical ML.

RAG & Retrieval: Retrieval-Augmented Generation (Hybrid Search, Re-ranking), Semantic Search, LlamaIndex, Vector DBs (Qdrant, PGVector).

Evaluation & LLM Ops: LLM-as-Judge, RAGAS (faithfulness, answer-relevancy), Precision / Recall / AUC, CI/CD Eval Gates, Tracing & Eval (Langfuse, OpenTelemetry), SigNoz, ELK Stack, CloudWatch, Prometheus.

Languages: Python, Java, TypeScript, SQL.

Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, FastAPI, Spring Boot, Django.

Cloud & Infra: Azure OpenAI, AWS (SageMaker, Bedrock, Lambda, ECS), Kubernetes, Docker, Terraform, Temporal.io.

Data Engineering: Snowflake Cortex, Apache Airflow, Kafka, Elasticsearch, ETL Pipelines, MongoDB, Redis, PostgreSQL, Browserbase.

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Senior AI/ML Engineer, Lead Architect

Remote

04/2025 – Present

- As **Lead Architect**, drove cloud-native GenAI delivery across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, translating ambiguous business problems into scalable, reliable production AI architectures.
- **Multimodal Document Intelligence (VTS):** Built an agentic **document-intelligence** pipeline that abstracts structured terms from commercial-lease documents with **structured-output validation**. Diagnosed a **10-point accuracy regression** and redesigned prompt strategy and **multimodal image preprocessing** to recover the full drop, restoring field-level extraction to its **~88% baseline** and unblocking downstream automation.
- **Field-Aware Extraction Engine (Outcomelabs.ai):** Architected an extraction platform processing **2M+ records/month** with a **"Strategy Router"** that classifies each field by complexity and routes ~60% to deterministic parsers, reserving the LLM for the rest, cutting token spend **~40%** with no measurable accuracy loss.
- **Agentic Automation over MCP:** Orchestrated a serverless **multi-agent** framework with **tool / function calling** over MCP (**Model Context Protocol**) and **swarm-based orchestration** for Salesforce automation. Cut idle compute via a pure **pay-per-execution** model, saving **~\$45k/year**.
- **RAG Search & Ranking:** Built a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5** by **~15 points (0.71 to 0.86)**.
- **Entity Matching & Evaluation:** Built an entity-matching pipeline on 22M records using an ensemble of **TF-IDF / Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier (85.3% AUC)**.
- **Scale & Impact:** Scaled the platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while optimizing fine-tuning strategies for cost efficiency.

Kuttl (Incubated at IIM Lucknow)

Founding Engineer

Remote

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet. Documented and standardized the codebase and infrastructure to ensure a **zero-downtime handover** to the incoming engineering team.
- **Fine-tuned Llama-3-8B** (LoRA / PEFT) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** while reducing per-token inference cost by **90%** and latency by 300ms.

- Operationalized the ML lifecycle with **CI/CD evaluation gates** using **LLM-as-judge** graders (**Ragas**) scoring **faithfulness and answer-relevancy**, automatically blocking deployments below threshold and catching quality regressions before production.
- Engineered a distributed document-parsing pipeline with **FastAPI** and **Temporal.io**, processing 50k+ daily messages with resilient retry logic and dead-letter queues to ensure no data loss.
- Applied semantic caching and request batching for RAG endpoints, improving **P99 latency by 40%** during high-concurrency bursts.
- Built the core backend and AI infrastructure from scratch, scaling the platform under strict startup cost constraints.

Amplicomm

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity **~42%**.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC (Role-Based Access Control)** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to use **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by **~40%** to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon "On the Hero Road" Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.